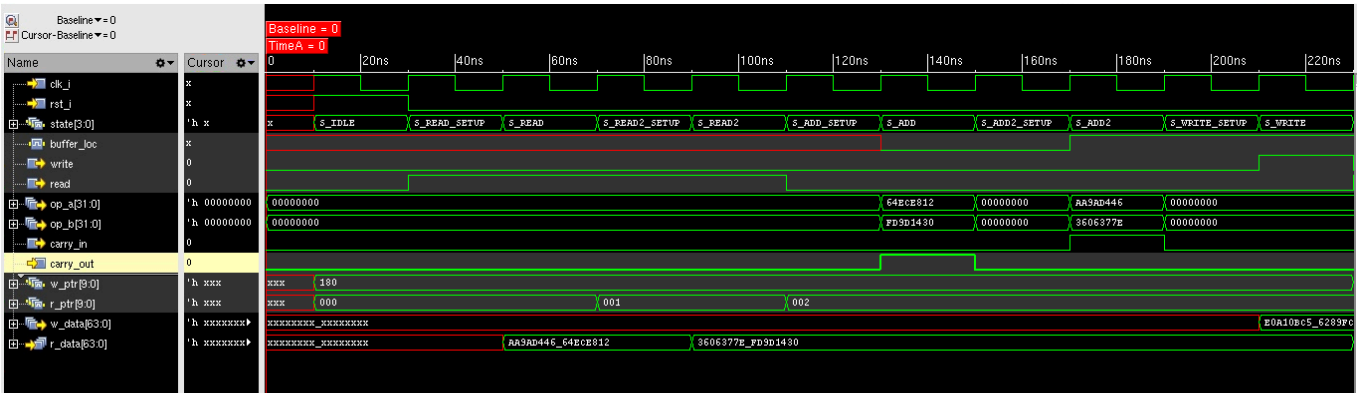
Kien Le's Submission (gatech-kien6, disc-kienlol.)

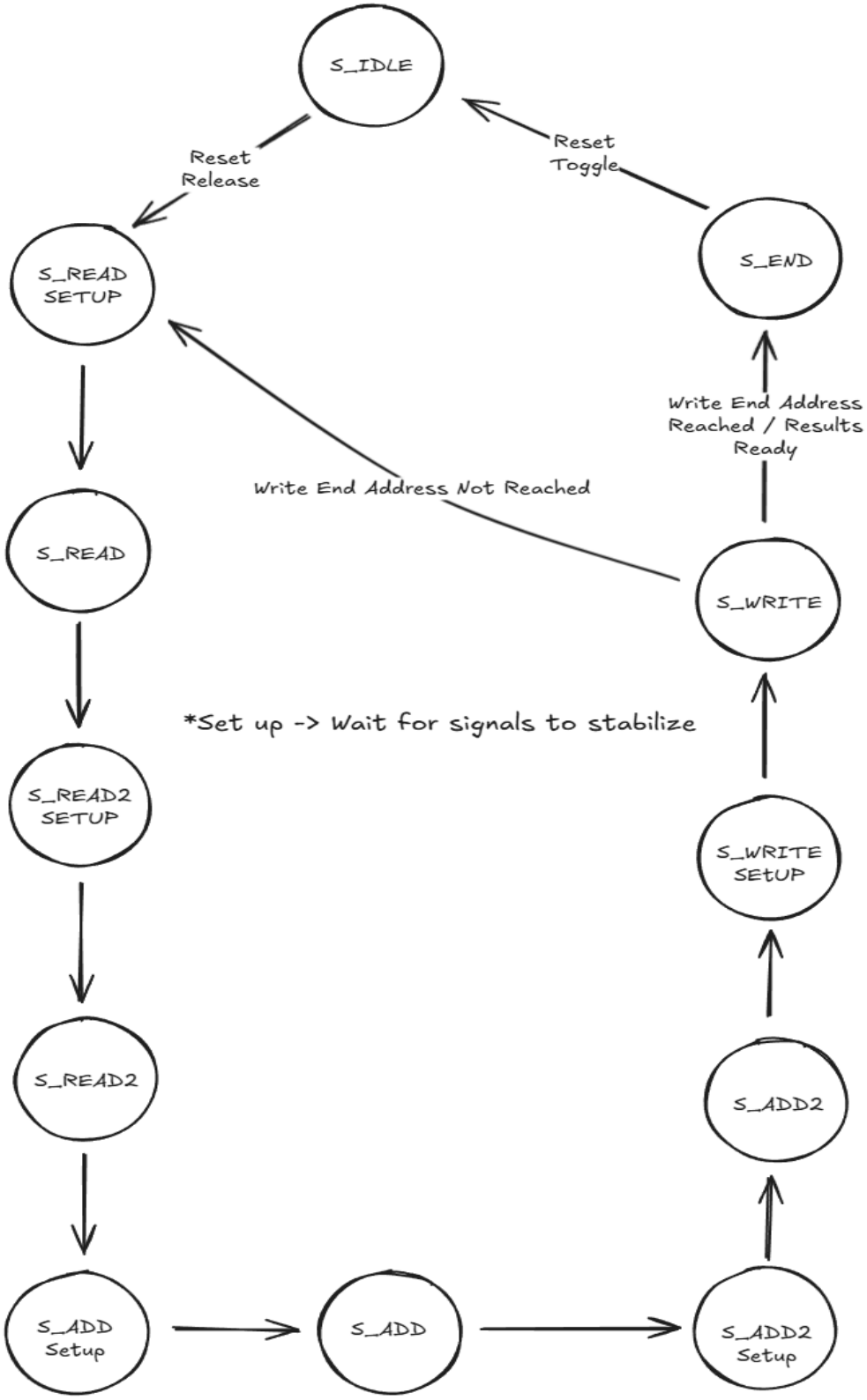
Passing Submission

```
kien6@ece-rschsrv:~
                        streams:  93, words: 63467
    logic [31:0] expected_post_lower [0:1023];
                                 |
xmelab: *W,COVMDD (./sym_links/tb_calculator.sv,23|35): Toggle coverage for bit, logic, reg, wire, enum and struct multi-dimensional static arrays and vectors is not supported by default. To enable toggle coverage for enum multi-dimensi
onal static arrays specify 'set_toggle_scoring -sv_enum enable_mda' and for other multi-dimensional static arrays, specify 'set_toggle_scoring -sv_mda [<max_bit_base2_exponent>] [-sv_mda_of_struct]' ccf command in the coverage configura
tion file.
xmelab: *W,COVNOEN: By default, toggle coverage is not supported for systemverilog enumerated nets and variables. To enable toggle coverage of these objects, specify the 'set_toggle_scoring -sv_enum' command in the coverage configuratio
n file.
        Building instance specific data structures.
        Loading native compiled code:     ................... Done
        Design hierarchy summary:
                                 Instances  Unique
                Modules:                41       8
                Verilog packages:        1       1
                Primitives:              6       2
                Registers:             253     125
                Scalar wires:          426       -
                Expanded wires:         64       2
                Vectored wires:         42       -
                Named events:            8       4
                Always blocks:          51      28
                Initial blocks:         16       9
                Parallel blocks:         1       1
                Cont. assignments:     168      84
                Pseudo assignments:    160       -
                Simulation timescale:  1ps
        Writing initial simulation snapshot: worklib.tb_calculator:sv
Loading snapshot worklib.tb_calculator:sv ..................... Done
xcelium> source /tools/software/cadence/xcelium/latest/tools/xcelium/files/xmsimrc
xcelium> run
xmsim: *W,SHMPOPT: Some objects excluded from $shm_probe due to optimizations.
           File: ./sym_links/tb_calculator.sv, line = 47, pos = 17
          Scope: tb_calculator
           Time: 0 FS + 0


--------------Beginning Simulation!--------------

Time:                    0
--------------Initializing Signals---------------

Time:                10000

-------------Finished Simulation!----------------

Time:            25730000
Cycle Count: 1285 cycles (from S_IDLE to S_END)
Simulation complete via $finish(1) at time 25730 NS + 0
./sym_links/tb_calculator.sv:65        $finish;
xcelium> exit

coverage setup:
  workdir  :  ./cov_work
  dutinst  :  tb_calculator(tb_calculator)
  scope    :  scope
  testname :  test

coverage files:
  model(design data) :  ./cov_work/scope/icc_2b0cbe78_00000000.ucm
  data               :  ./cov_work/scope/test/icc_2b0cbe78_00000000.ucd
TOOL:   xrun(64)       25.03-s003: Exiting on Feb 08, 2026 at 17:11:10 EST  (total: 00:00:01)
make[1]: Leaving directory '/nethome/kien6/Spring26-DD-Onboarding/sim/behav'
python3.12 ../../scripts/check_onboarding.py /nethome/kien6/Spring26-DD-Onboarding/sim/behav/WORKSPACE/memory_post_state_lower.txt /nethome/kien6/Spring26-DD-Onboarding/sim/behav/WORKSPACE/sim_memory_post_state_lower.txt
PASSED: RTL simulation output matches expected results.
python3.12 ../../scripts/check_onboarding.py /nethome/kien6/Spring26-DD-Onboarding/sim/behav/WORKSPACE/memory_post_state_upper.txt /nethome/kien6/Spring26-DD-Onboarding/sim/behav/WORKSPACE/sim_memory_post_state_upper.txt
PASSED: RTL simulation output matches expected results.
kien6@ece-rschsrv.ece.gatech.edu>
```

Waveform for a Cycle of Addition (0:230ns)



FSM Implementation



Write-Up

For my implementation of the FSM, I knew I had to incorporate states within to the diagram that would allow the signals to stabilize as working with unstable values would just be a pain to debug, so I initially started with a lot more setup states that I basically used as a wait to allow time for the values to stabilize. Instead of the one setup stages that I currently have, I had about two per each major instruction (read, add, and write) just to be safe. From there, once I had setup the logic for the controller and could use the Simvision tool to visualize the actual signals and when they were available, I could cut back on the number of states that I implemented to reduce clock cycles.
Working with additional states to wait allowed me to experiment with the sequence of which I could raise flags and access values without messing up the timing in contrast to setting up a limited amount of wait states and dealing with timing and access conflicts.

Walking through the cycle of addition, I follow the implementation described, reading in the 64-bit values to operands a and b before then splicing them into the lower and upper 32 bit in the READ states, incrementing the read pointers to keep track of the value that had just been read. A wait/setup state was incorporated in between these READs to allow for the r_ptr to stabilize to ensure that the right value is read in subsequent state. From there, I setup for the ADD states with the first one focused on lower bits of op_a and op_b using the initial carry_in value of 0 (no previous values calculation). Then I need to keep track of the carry_out with a logic variable that can be passed on to the following calculations for the upper bits. The tracked carry_out value then propagates to ADD2 where the upper bits of op_a and op_b are calculated and stored into an upper result that is 32 bits. The carry_out for this operation can be discarded since there are no other calculations for any higher bits that would be dependent on this value. Write_setup then concetenates the calculated values of the upper and lower together and stores it to the w_data before write increments to keep track of where it was written. This entire process will terminate when the write end address is reached, otherwise it will continually loop back to the read_setup and start again.